ISHRAT BHULLAR

AI Engineer · Retrieval-Augmented Generation & Backend Systems

ibhullar_be22@thapar.edu | linkedin.com/in/ishrat-bhullar-4a1441295 | github.com/Ishrat-Bhullar | Patiala, India

SUMMARY

AI Engineer specialising in retrieval-augmented generation and LLM-powered applications, with delivery experience from an AI engineering internship at **EY** and two self-directed systems builds. Builds question-answering systems that return accurate, source-traceable answers over private document collections, including scanned and non-English material. Strongest in hybrid retrieval, vector search, and FastAPI backends, with experience deploying LLMs entirely on-premise for clients who cannot send data to external APIs.

EXPERIENCE

AI Engineering Intern — EY (Ernst & Young) — May 2026 – Jul 2026

Full-stack contributor on a small AI engineering team delivering a GenAI proof-of-concept for a government client.

Python · FastAPI · React · PostgreSQL · ChromaDB · FAISS · BM25 · Ollama · Docker

- Built a **hybrid retrieval engine** that merges FAISS dense search with BM25 lexical scoring through **Reciprocal Rank Fusion**, recovering exact-match terms dense-only search missed.
- Automated knowledge-base refresh with an **OCR ingestion pipeline** that converts scanned documents into chunked, metadata-tagged embeddings, replacing a manual update process.
- Extended retrieval to non-English queries by integrating **multilingual E5 embeddings**, letting a single index serve cross-lingual search without separate per-language stores.
- Served retrieval, inference, and ingestion through FastAPI REST endpoints, connecting a React chat interface to a PostgreSQL store for conversations and document metadata.
- Deployed open-source LLMs on-premise through **Ollama** to satisfy client data-residency rules, keeping document content inside the client network and removing external API dependency.
- Cut query latency by caching embeddings for repeat lookups, batching inference calls, and tuning vector index parameters against the client's document set.
- Grounded responses in retrieved text through inline citation generation, mapping each claim to its source document and page so answers stay traceable.

PROJECTS

Multi-Agent Autonomous SDLC Platform — Independent Project — 2026

FastAPI · React · PostgreSQL · LangGraph · LLM Agents

- Architected a **nine-agent pipeline** that turns a plain-language brief into code, covering requirements, architecture, database design, generation, review, security, and documentation.
- Sequenced agent handoffs through a stateful orchestrator with **human approval gates**, so a reviewer can edit or reject any agent's output before the next stage runs.
- Surfaced agent status, generated artifacts, and approval history in a React dashboard, persisting full run state in PostgreSQL so runs stay auditable.

Autonomous Water-Cleaning Surface Vehicle — Academic Project — 2025

Arduino · Embedded C · IR & Ultrasonic Sensors · IoT · Blockchain

- Engineered an Arduino-controlled surface vehicle that navigates autonomously and triggers filtration, combining IR and ultrasonic input for obstacle and contamination detection.
- Logged sensor telemetry to a tamper-evident blockchain ledger, and exposed live device status, cycle history, and threshold alerts through a web dashboard.

GPU-Accelerated Stock Forecasting — Academic Project — 2025

Python · CUDA · NumPy · Pandas · Parallel Computing

- Accelerated a time-series forecasting model on historical equity data by moving feature computation and training onto the GPU with **CUDA**.
- Benchmarked GPU against CPU runtimes across increasing data volumes, profiling host-to-device transfer overhead to tune batch size and kernel launch configuration.

TECHNICAL SKILLS

AI & Retrieval	Retrieval-Augmented Generation (RAG), hybrid dense + sparse retrieval, BM25, Reciprocal Rank Fusion, vector search, embeddings (multilingual E5), chunking strategies, prompt engineering, citation grounding, OCR ingestion pipelines, LLM agents
Backend & Data	Python, FastAPI, REST API design, PostgreSQL, SQL, ChromaDB, FAISS
Model Serving	Ollama (local LLM deployment), on-premise inference, embedding caching, batched inference
Tools	Git, GitHub, Docker, Linux
Working Knowledge	React, LangGraph, CUDA / GPU computing, C++, NumPy, Pandas
Coursework	Data Structures & Algorithms, Operating Systems, Computer Networks, DBMS, Software Engineering

EDUCATION

B.E. in Computer Engineering — Thapar Institute of Engineering & Technology, Patiala — 2022 – 2026

CGPA 7.58 / 10 · Sacred Heart Sr. Sec. School, Chandigarh — CBSE Class XII (Non-Medical): 93.4%

LEADERSHIP & INVOLVEMENT

- Core team member across five student organizations; 2 years as graphic designer for **Echoes**, the official university magazine.
- **HSBC Indian Business Case Program** · **UNICEF India U-Report** advocate · **NSS** — 120+ volunteer hours.